

Armaan Dhah

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EDUCATION

University of California, Los Angeles | Los Angeles, CA

Sep. 2025 – June 2027

B.S. Mechanical Engineering; GPA: 3.7/4.0

Clovis Community College | Fresno, CA

Aug. 2023 – May 2025

Mechanical Engineering; GPA: 4.0/4.0

SKILLS

Manufacturing: 3D Printing, Lathe, Mill, CNC Mill, Soldering, TIG Welding, Spot Welding, Drill Press, Angle Grinder, Bandsaw

Software: SolidWorks, FEA (ANSYS & SolidWorks), Fusion CAM, Altium Co-Designer, MATLAB, VBA, AutoCAD, Microsoft Suite

Certifications: Certified SolidWorks Associate (CSWA)

WORK EXPERIENCE

Advanced Spade Company

May 2026 – Sept. 2026

Mechanical Engineering Intern

- Redesigned a pitch/roll camera gimbal for a sewer-inspection robot, cutting package volume 53% through a re-tuned gear ratio, a ball-bearing-supported roll shaft, and slip-ring integration, with Teensy 4.1 control.
- Led mechanical design across three phases of an in-house GNSS positioning device, from a retrofitted test fixture through a complete four-board production unit, including a custom 2s2p battery pack.
- Redesigned the external battery charger enclosure, built a T-slot Testbench Mobility Unit cart, and maintained an automated 3D-printing farm across the product line.

Clovis Community College Tutorial Center | Fresno, CA

Aug. 2024 – May 2025

Team Lead – Student Aide II

- Provided math, physics, chemistry, and engineering tutoring, supporting students in developing problem-solving skills and conceptual understanding.

ENGINEERING PROJECTS

UCLA Bruin Baja SAE | Los Angeles, CA

Suspension Lead Engineer

May 2026 – Present

- Direct a team of 13 project engineers through the design, analysis, fabrication, and validation of the entire suspension and steering system, coordinating directly with other subsystem leads to ensure flawless vehicle integration.
- Optimize suspension geometry to elevate vehicle handling and reliability while supervising component packaging, mass reduction, and strict rule compliance.
- Manage the subteam budget and defend design architectures at formal technical reviews and competition design presentations.

Suspension Team Engineer

Sept. 2025 – May 2026

- Fabricated upper and lower A-arms via angle-grinder tube notching, DXF templates, and dedicated jiggling, and machined tightly toleranced spacers and bushings on the manual mill and lathe.
- Manufactured a limit switch plate for the centerlock in Fusion CAM, and helped redesign a critical A-arm after a late-season failure, 3D printing test models to validate the new geometry.

Flagship Combat Robotics – UCLA ASME | Los Angeles, CA

Sept. 2025 – May 2026

SKULL CRUSHER, Weapon Engineer

- Optimized the weapon geometry of a 30 lb combat robot for weight while meeting hit targets, and machined the hub spline, weapon shaft, motor shaft, and pulleys on manual and CNC mills, programming the hub spline toolpaths in Fusion CAM.
- Ran ANSYS FEA across design revisions to validate spline hub survivability, and designed the system for on-the-fly maintenance so parts could be swapped between fights.

UCLA Combat Robotics | Los Angeles, CA

May 2026

U.A.A.O.D.A.D., Lead Engineer

- Designed and iterated a parameterized eggbeater weapon geometry for a 3 lb combat robot with a teammate in one week, using a weight-versus-energy sweep to size the disc within a 0.7 lb target while clearing a 300 J minimum.
- Machined the weapon disc and drivetrain on manual and CNC mills, harnessed and soldered the full electrical system under a 12-hour pre-competition crunch, and competed with the finished robot.

Crush Depth MATE ROV Underwater Robotics | Fresno, CA

Aug. 2024 – June 2025

FLOAT Lead Mechanical Engineer

- Led the mechanical design and fabrication of an underwater vertical profiling float's buoyancy engine, structure, and waterproofing, iterating eight buoyancy and housing assemblies in SolidWorks.
- Engineered an O-ring waterproofing interface depth-tested to 3.0 meters with zero leakage over 150+ hours, a 90% improvement over previous years.
- Supervised teammates in CAD modeling, fabrication, and mechanical testing, ensuring milestone completion under tight competition deadlines, and submitted the system to the 2025 MATE ROV Explorer Class Championship.